第10讲笔记

定量数据分析

引言

- 我们已经开始检查模型的设定方式。
- 上一讲我们认为，半对数模型优于最初的线性模型。
- 本讲将进一步改进模型，并讨论分析所依赖的其他假设。

模型假设

我们将讨论以下假设（引自Peter Kennedy的《计量经济学指南》）：

1. 模型设定正确。
2. 残差方差相同，并且彼此独立。
3. 自变量之间不存在完全线性关系。
4. 自变量是固定的，且测量没有误差。
5. 残差的期望为零。

改进我们的模型

- 我希望开始从模型中移除一些无关变量，使模型更容易处理。
- 但很难判断什么是无关变量，因为我们知道：如果模型设定错误，就不能相信p值。
- 因此这将是一个迭代过程。
- 这也意味着，我们必须运用领域知识、理论和已有研究来指导这一过程。

假设3

- 线性模型的第三项假设，也可能让我们难以判断哪些变量显著。
- 正式表述是：自变量之间不存在完全线性关系。
- 从这个角度看，假设本身没有问题。
- 一旦违反这个假设，模型就无法正常估计。
- 它的意思是：任何一个自变量都不能由其余自变量的某种线性组合完全推导出来。

例子

- 假设研究食物贫困，想知道收入对家庭食品支出的影响。

- 假设某个家庭有两名成年人，两人都有收入。
- 模型中有三个自变量：

1. 成员A的收入。
2. 成员B的收入。
3. 家庭总收入。

- 这违反了模型假设，因为家庭总收入就是成员A和成员B收入之和。
- 如果尝试估计该模型，R会自动剔除其中一个变量。
- 这个问题叫作完全多重共线性。

为什么会这样?

- 如果想知道它为何使模型无法估计，可以尝试求解模型背后的线性代数。
- 最终会遇到除以零的操作，而这是不可能的。
- 不过，从直觉上也能理解为什么无法分别估计每个变量的效应。
- 例如，不可能观察到两名成员收入都高、但家庭总收入却低的家庭。
- 也不可能观察到两人收入都低、但家庭总收入却高的家庭。
- 缺少这样的独立变化，就无法估计每个变量各自的贡献。

程度问题

- 只有当所尝试的模型本身在逻辑上不成立时，才会严格违反该假设。
- 然而，这也是一个程度问题。
- 只要自变量之间存在相关性，这个问题就可能在一定程度上存在。
- 相关性越高，多重共线性问题越严重。

多重共线性

- 当多重共线性程度很高时，可能出现一系列问题。
- 标准误可能变大，导致估计效率降低，也更难发现统计上显著的系数。
- 某些变量的回归系数也可能出现不符合预期的符号。
- 换用不同模型设定后，系数估计可能大幅波动；例如，加入或删除变量就会导致较大变化。

识别多重共线性

- 请记住，这是程度问题。
- 我们可能事先就怀疑自变量间存在相关性。

- 也可能观察到模型出现异常行为，例如：
- 标准误很大，尤其是在本不应如此的时候。
- 模型中的回归系数不稳定。
- 可以查看自变量之间的两两相关系数。
- 还可以使用方差膨胀因子等更复杂的指标。
- 只要多重共线性没有真正影响分析，它就不是问题。

方差膨胀因子（VIF）

- 方差膨胀因子的思想是：查看某个自变量的变异有多少能由其他自变量解释。
- VIF表示多重共线性使系数估计方差膨胀了多少。
- VIF为1表示没有多重共线性。
- 低于5通常不用担心。
- 超过10则可能需要留意。
- 只要没有出现相关症状，就未必构成问题。
- 可以使用car包中的vif()函数计算（广义）方差膨胀因子。

模型的VIF值

```
lfs %>% mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 +
MARSTA + INDE07M + NSECMJ20 + GOR9D,
     data=.) %>%
 vif()
```

```
##              GVIF Df GVIF^(1/(2*Df))
## SEX      1.249367  1        1.117751
## AGE      1.682555  1        1.297133
## HIQUL22D 1.609476  5        1.048741
## FLEX22W6 1.124745  1        1.060540
## FTPT     1.266411  1        1.125349
## EMPLEN   1.431578  7        1.025958
## JOBTYP   1.122614  1        1.059535
## MPNR02   1.269364  8        1.015019
## MARSTA   1.432800  5        1.036618
## INDE07M  1.668301  8        1.032505
## NSECMJ20 2.151890  6        1.065946
## GOR9D    1.190306 11        1.007950
```

结果解释

- 当模型包含因子变量时，通常建议使用缩放后的GVIF值（GVIF^(1/(2*Df))）。
- 这里没有迹象表明多重共线性构成问题。
- 因此不应看到由多重共线性引起的标准误膨胀。

如何处理多重共线性?

- 首先应问：它真的需要处理吗？
- 可以考虑删除某个变量；但如果因此造成遗漏变量问题，就会得不偿失。
- 还可考虑降维，例如因子分析或主成分分析。
- 但这样会让结果更难解释。
- 使用联合假设检验：检验一组系数，而不是逐个检验。
- 也有人建议使用岭回归，但这种方法给出有偏估计。

假设4

- 第四项假设对自变量提出了一些要求。
- 其中一部分是要求自变量具有外生性。
- 这实际上是一个庞大的话题，我们尽量简化讨论。
- 必须特别注意回归方程左右两侧是否出现同一个变量，例如将身体质量指数表示为身高的函数。
- 必须避免自变量可能受到因变量影响的情况。

假设4与我们的模型

- 诊断这些问题非常复杂。
- 最简单的做法是认真反思模型，并参考已有研究和理论。
- 我们的回归方程没有把工资同时放在等号左右两边。
- 但是，是否有不具外生性的自变量？例如，哪些自变量可能反过来由工资决定？

我们的自变量

- 年龄
- 全职 / 兼职
- 零工时合同
- 永久性工作
- 地区

- 在当前雇主处工作的时间
- 教育
- 行业
- 工作类型
- 婚姻状况
- 企业规模

问题

- 我认为，在当前雇主处工作的时间可能有问题。
- 我们的初衷是使用在现雇主处工作的时间来衡量就业经验。
- 工作经验越丰富，工资可能越高。
- 然而，如果当前雇主支付的工资很高，我可能会在那里工作更久。
- 如果工资较低，我可能很快离职。
- 所以模型中可能存在内生关系。

解决办法

- 处理内生变量可能非常复杂。
- 我建议尽量避免设定存在此类关系的模型。
- 对于当前模型，我建议将这个变量排除。
- 因为无法确定相关结果能否得到可靠解释。
- 这既涉及回归系数，也涉及标准误。

改进我们的模型

- 回到回归模型。
- 首先回顾上一讲的结果。
- 从线性模型开始。

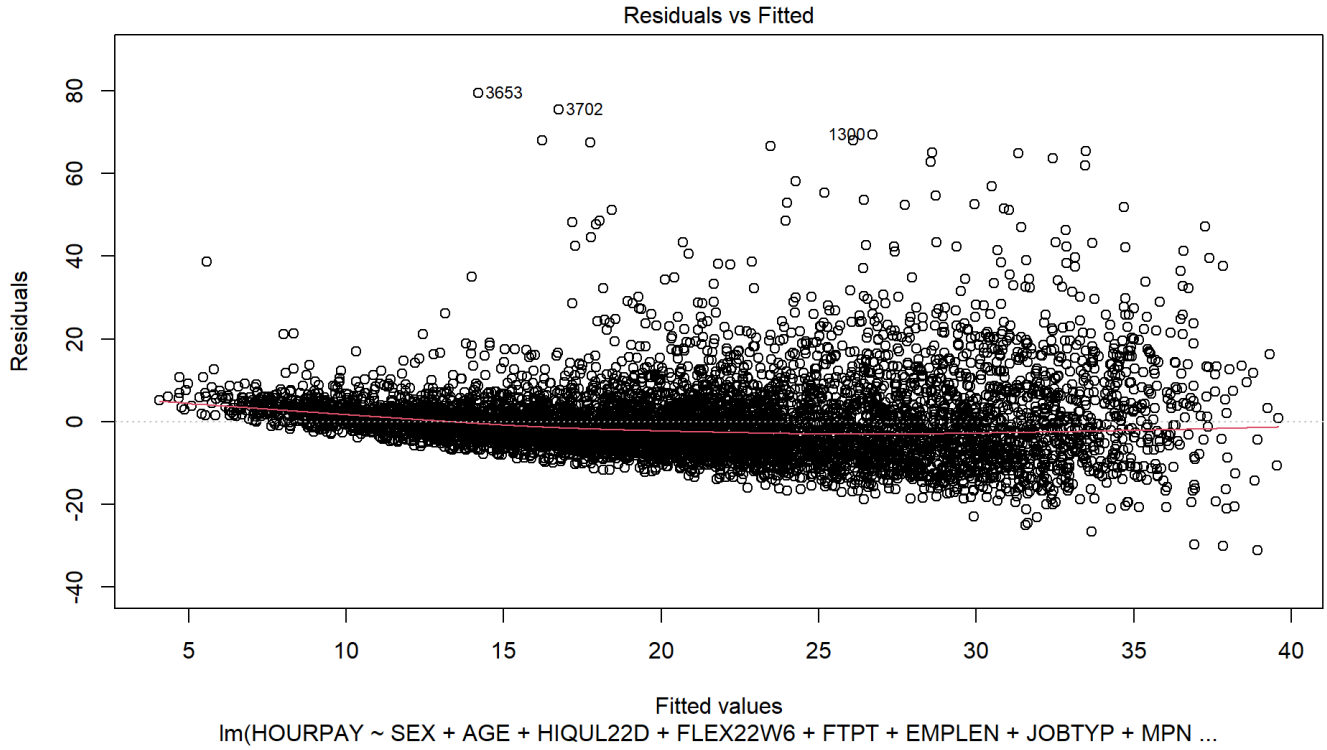
原始线性模型

```
##
## Call:
## lm(formula = HOURPAY ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT +
##     EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 +
##     GOR9D, data = .)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -31.140  -5.101  -1.189   3.046  79.547
##
## Coefficients:
##                                                    Estimate Std. Error t value
## (Intercept)                                        25.78853    1.98215  13.010
## SEXFemale                                          -1.78189    0.23677  -7.526
## AGE                                                 0.10231    0.01142   8.956
## HIQUL22DHigher education                           -3.23076    0.42198  -7.656
## HIQUL22DA level or equivalent                      -3.85940    0.29611 -13.034
## HIQUL22DGCSE A*-C or equivalent                    -4.53640    0.34719 -13.066
## HIQUL22DOther qualitfication                       -4.44594    0.55803  -7.967
## HIQUL22DNo Qualification                           -6.14744    0.68828  -8.932
## FLEX22W6No                                          0.40302    0.78712   0.512
## FTPTPart-time                                      -0.07943    0.28902  -0.275
## EMPLENThree months, less than 6                    -0.39861    0.85519  -0.466
## EMPLEN6 months, less than 12                       -0.18981    0.78648  -0.241
## EMPLEN1 year, less than 2                          -0.30792    0.71735  -0.429
## EMPLEN2 years, less than 5                          0.34445    0.68859   0.500
## EMPLEN5 years, less than 10                         0.52264    0.70371   0.743
## EMPLEN10 years, less than 20                        1.06104    0.70594   1.503
## EMPLEN20 years or more                              1.57930    0.73389   2.152
## JOBTYPNot permanent in some way                     0.15802    0.62841   0.251
## MPNR0211-19                                         0.01830    0.46778   0.039
## MPNR0220-24                                         0.17017    0.60750   0.280
## MPNR02Don't know but under 25                      -1.01501    0.78914  -1.286
## MPNR0225-49                                         0.05431    0.41028   0.132
## MPNR0250-249                                        0.51390    0.35217   1.459
## MPNR02250-499                                       1.50815    0.50016   3.015
## MPNR02Don't know but between 50 and 499             0.65691    0.58493   1.123
## MPNR02500 or more                                   2.40393    0.35823   6.711
## MARSTAMarried, living with spouse                   1.44619    0.25735   5.620
## MARSTAMarried separated from spouse                 0.79898    0.73767   1.083
## MARSTADivorced                                      1.68799    0.46047   3.666
## MARSTAWidowed                                       1.62508    1.09137   1.489
## MARSTACurrently or previously in civil partnership -0.38073    1.05538  -0.361
## INDE07MEnergy and water                             3.70456    1.74382   2.124
## INDE07MManufacturing                                2.19235    1.62412   1.350
## INDE07MConstruction                                 3.84059    1.65911   2.315
## INDE07MDistribution, hotels and restaurants        -0.39967    1.60802  -0.249
## INDE07MTransport and communication                  3.64347    1.62184   2.246
```

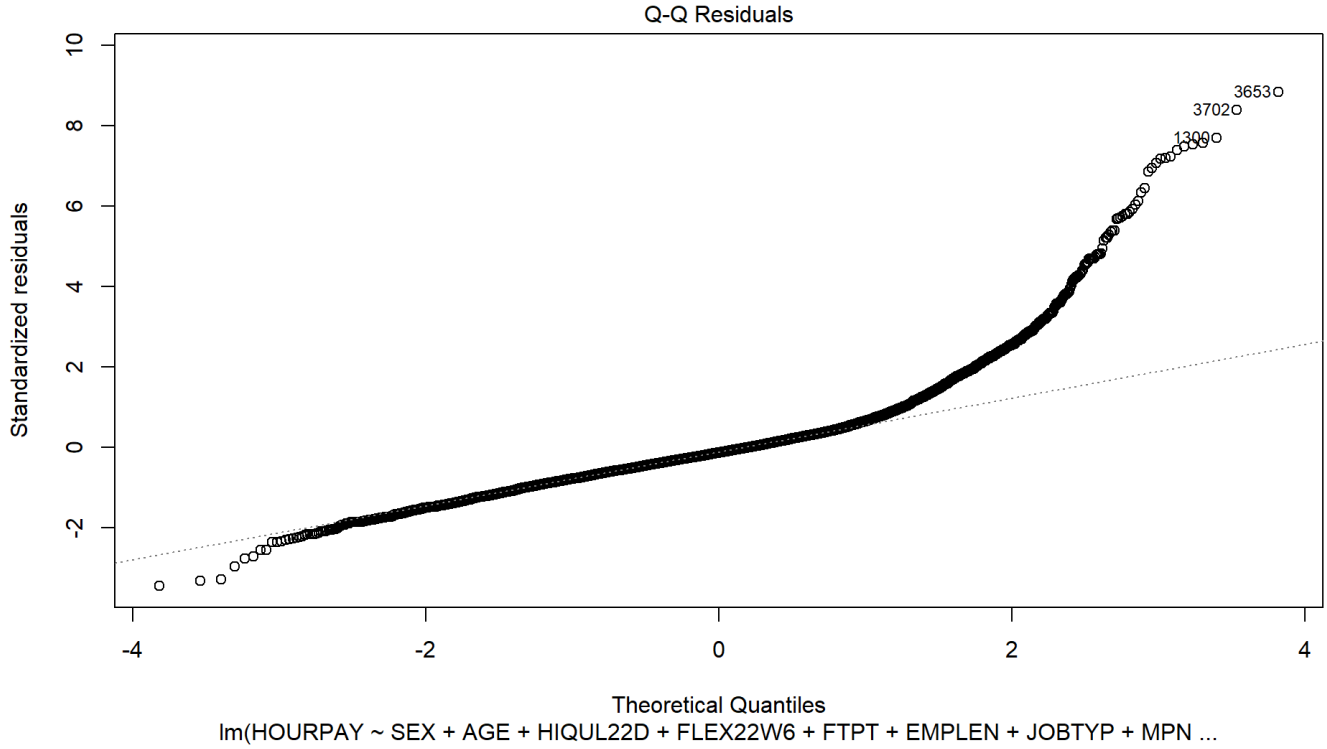
```
## INDE07MBanking and finance                          2.61233    1.60432   1.628
## INDE07MPublic admin, education and health           -0.37672    1.59701  -0.236
## INDE07MOther services                               -0.61520    1.66640  -0.369
## NSECMJ20Lower managerial and professional           -5.89892    0.29352 -20.097
## NSECMJ20Intermediate occupations                   -10.52187    0.37006 -28.433
## NSECMJ20Lower supervisory and technical             -9.81153    0.51984 -18.874
## NSECMJ20Semi-routine occupations                   -11.54158    0.45036 -25.627
## NSECMJ20Routine occupations                        -12.49647    0.47149 -26.504
## NSECMJ20Never worked                                -8.66825    0.81832 -10.593
## GOR9DYorkshire and The Humber                       -4.31942    0.50909  -8.485
## GOR9DSouth West                                     -3.69832    0.50952  -7.258
## GOR9DWest Midlands                                  -3.25102    0.54650  -5.949
## GOR9DSouth East                                     -2.42283    0.47010  -5.154
## GOR9DEast of England                                -2.23360    0.53480  -4.176
## GOR9DEast Midlands                                  -4.01808    0.52771  -7.614
## GOR9DNorth East                                     -5.24307    0.65972  -7.947
## GOR9DNorth West                                     -4.00454    0.52126  -7.682
## GOR9DWales                                          -4.97962    0.66622  -7.474
## GOR9DNorthern Ireland                               -5.08488    0.57151  -8.897
## GOR9DScotland                                       -3.28323    0.53089  -6.184
##                                                    Pr(>|t|)
## (Intercept)                                         < 2e-16 ***
## SEXFemale                                          5.87e-14 ***
## AGE                                                 < 2e-16 ***
## HIQUL22DHigher education                           2.16e-14 ***
## HIQUL22DA level or equivalent                       < 2e-16 ***
## HIQUL22DGCSE A*-C or equivalent                     < 2e-16 ***
## HIQUL22DOther qualitfication                       1.87e-15 ***
## HIQUL22DNo Qualification                            < 2e-16 ***
## FLEX22W6No                                         0.608648
## FTPTPart-time                                      0.783459
## EMPLENThree months, less than 6                    0.641152
## EMPLEN6 months, less than 12                       0.809302
## EMPLEN1 year, less than 2                          0.667754
## EMPLEN2 years, less than 5                         0.616933
## EMPLEN5 years, less than 10                        0.457692
## EMPLEN10 years, less than 20                       0.132879
## EMPLEN20 years or more                             0.031434 *
## JOBTYPNot permanent in some way                    0.801461
## MPNRO211-19                                        0.968800
## MPNRO220-24                                        0.779394
## MPNRO2Don't know but under 25                      0.198407
## MPNRO225-49                                        0.894685
## MPNRO250-249                                       0.144539
## MPNRO22250-499                                     0.002576 **
## MPNRO2Don't know but between 50 and 499            0.261452
## MPNRO2500 or more                                  2.08e-11 ***
## MARSTAMarried, living with spouse                  1.98e-08 ***
```

```
## MARSTAMarried separated from spouse                0.278795
## MARSTADivorced                                     0.000248 ***
## MARSTAWidowed                                      0.136521
## MARSTACurrently or previously in civil partnership 0.718297
## INDE07MEnergy and water                            0.033671 *
## INDE07MManufacturing                               0.177099
## INDE07MConstruction                                0.020649 *
## INDE07MDistribution, hotels and restaurants        0.803718
## INDE07MTransport and communication                 0.024702 *
## INDE07MBanking and finance                         0.103503
## INDE07MPublic admin, education and health          0.813522
## INDE07MOther services                              0.712006
## NSECMJ20Lower managerial and professional           < 2e-16 ***
## NSECMJ20Intermediate occupations                    < 2e-16 ***
## NSECMJ20Lower supervisory and technical             < 2e-16 ***
## NSECMJ20Semi-routine occupations                    < 2e-16 ***
## NSECMJ20Routine occupations                         < 2e-16 ***
## NSECMJ20Never worked                                < 2e-16 ***
## GOR9DYorkshire and The Humber                       < 2e-16 ***
## GOR9DSouth West                                    4.32e-13 ***
## GOR9DWest Midlands                                 2.83e-09 ***
## GOR9DSouth East                                    2.62e-07 ***
## GOR9DEast of England                               2.99e-05 ***
## GOR9DEast Midlands                                 2.99e-14 ***
## GOR9DNorth East                                    2.19e-15 ***
## GOR9DNorth West                                    1.77e-14 ***
## GOR9DWales                                         8.65e-14 ***
## GOR9DNorthern Ireland                               < 2e-16 ***
## GOR9DScotland                                      6.57e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.055 on 7291 degrees of freedom
## Multiple R-squared:  0.389,  Adjusted R-squared:  0.3844
## F-statistic: 84.41 on 55 and 7291 DF,  p-value: < 2.2e-16
```

残差与拟合值图



残差QQ图



RESET检验

```
lfs %>%
  lm(HOURPAY ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNRO2 +
MARSTA + INDE07M + NSECMJ20 + GOR9D, data=.) %>%
  resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 123.33, df1 = 2, df2 = 7289, p-value < 2.2e-16
```

BP检验

```
lfs %>%
  lm(HOURPAY ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNRO2 +
MARSTA + INDE07M + NSECMJ20 + GOR9D, data=.) %>%
  bptest()
```

```
##
##  studentized Breusch-Pagan test
##
## data:  .
## BP = 390.68, df = 55, p-value < 2.2e-16
```

半对数模型

- 当前模型没有满足相关假设。
- 上一讲通过对数变换改进了模型。

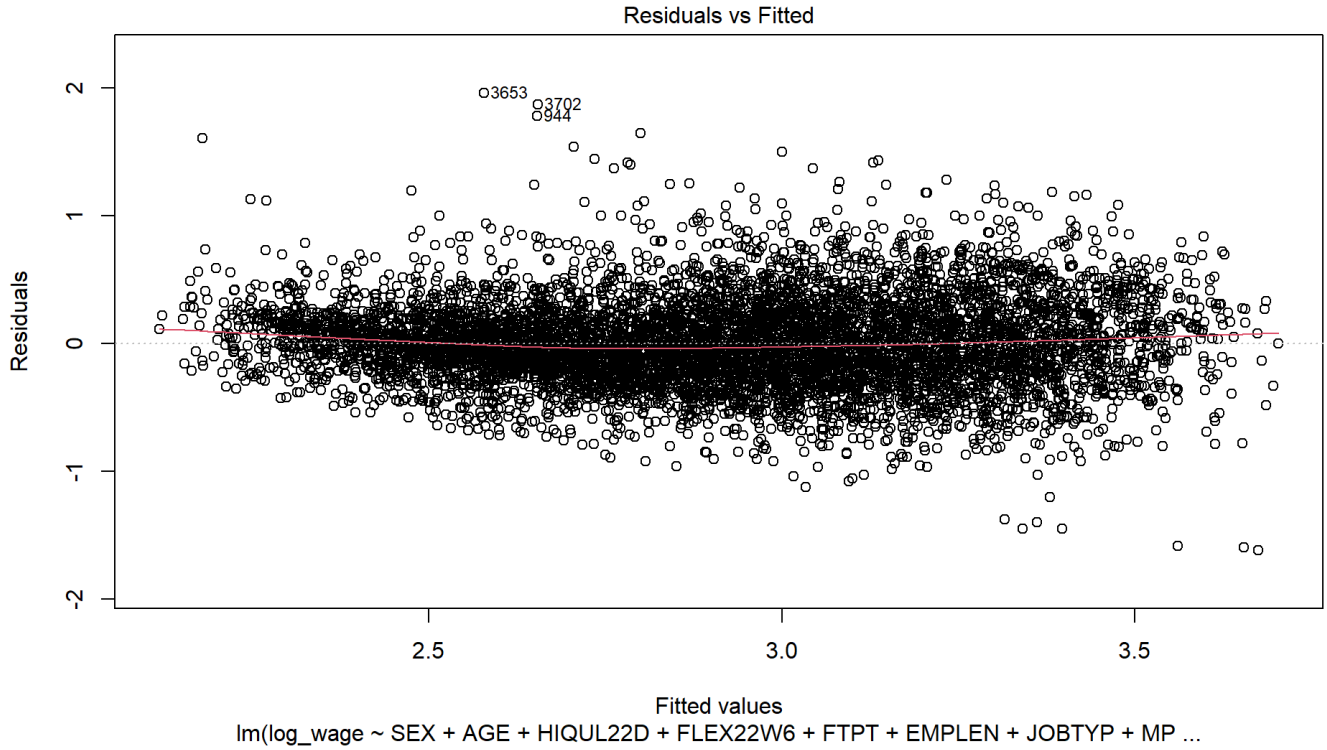
半对数线性模型

```
##
## Call:
## lm(formula = log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT +
##     EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 +
##     GOR9D, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.62055 -0.21109 -0.01861  0.19648  1.96285
##
## Coefficients:
##                                                    Estimate Std. Error
## (Intercept)                                       3.0115227  0.0744636
## SEXFemale                                        -0.0719738  0.0088949
## AGE                                               0.0041173  0.0004292
## HIQUL22DHigher education                         -0.1352929  0.0158525
## HIQUL22DA level or equivalent                    -0.1692705  0.0111238
## HIQUL22DGCSE A*-C or equivalent                  -0.2011436  0.0130428
## HIQUL22DOther qualitfication                     -0.1975554  0.0209636
## HIQUL22DNo Qualification                         -0.2873597  0.0258566
## FLEX22W6No                                        0.0567135  0.0295696
## FTPTPart-time                                    -0.0297625  0.0108578
## EMPLENThree months, less than 6                  -0.0151183  0.0321268
## EMPLEN6 months, less than 12                      0.0146695  0.0295457
## EMPLEN1 year, less than 2                         0.0065315  0.0269487
## EMPLEN2 years, less than 5                        0.0477627  0.0258684
## EMPLEN5 years, less than 10                       0.0642989  0.0264364
## EMPLEN10 years, less than 20                      0.0872838  0.0265202
## EMPLEN20 years or more                            0.1164016  0.0275700
## JOBTYPNot permanent in some way                   0.0104594  0.0236075
## MPNR0211-19                                       0.0111125  0.0175732
## MPNR0220-24                                       0.0178841  0.0228220
## MPNR02Don't know but under 25                    -0.0260251  0.0296457
## MPNR0225-49                                       0.0139941  0.0154129
## MPNR0250-249                                      0.0441019  0.0132299
## MPNR02250-499                                     0.0889228  0.0187895
## MPNR02Don't know but between 50 and 499           0.0365645  0.0219740
## MPNR02500 or more                                 0.1195901  0.0134575
## MARSTAMarried, living with spouse                 0.0579014  0.0096678
## MARSTAMarried separated from spouse               0.0358534  0.0277121
## MARSTADivorced                                    0.0675272  0.0172986
## MARSTAWidowed                                     0.0260218  0.0409994
## MARSTACurrently or previously in civil partnership  0.0012794  0.0396475
## INDE07MEnergy and water                           0.1886694  0.0655100
## INDE07MManufacturing                              0.1221503  0.0610133
## INDE07MConstruction                               0.1949621  0.0623280
## INDE07MDistribution, hotels and restaurants      -0.0073309  0.0604083
## INDE07MTransport and communication                0.1702397  0.0609278
```

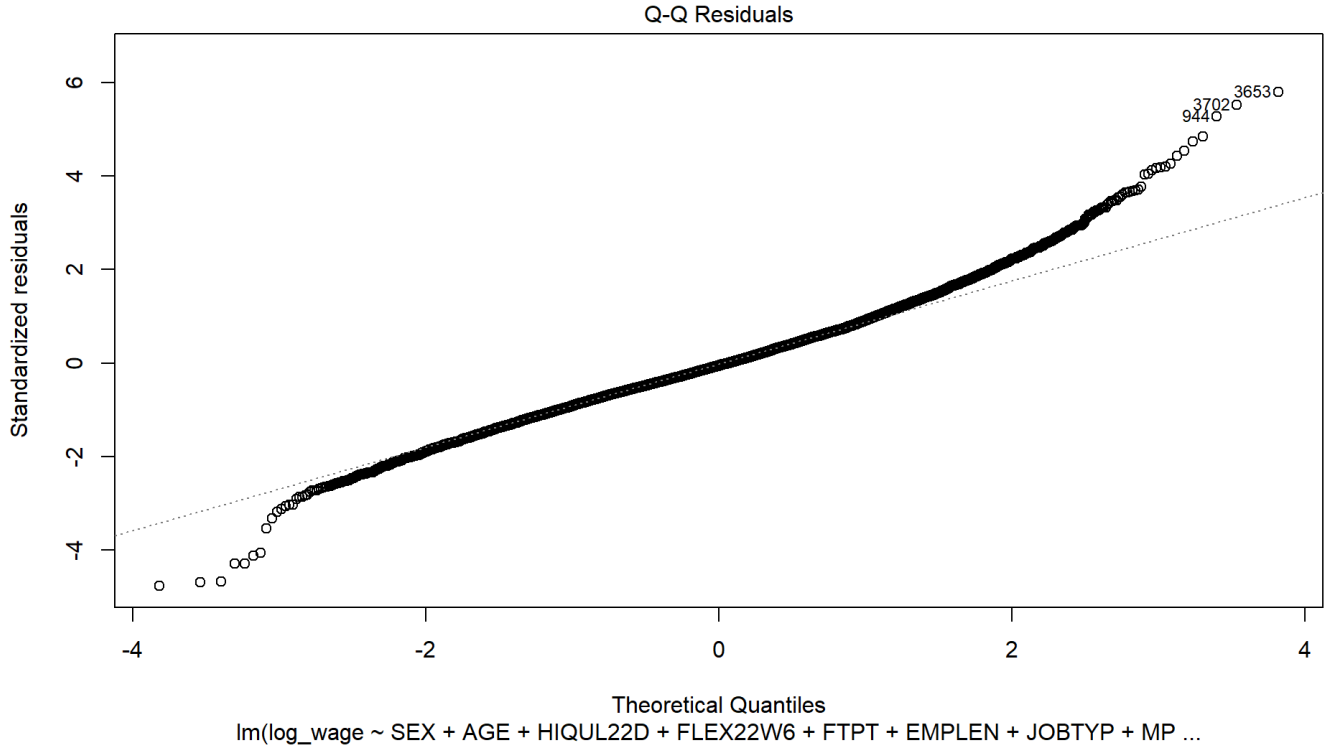
```
## INDE07MBanking and finance                          0.1231448  0.0602697
## INDE07MPublic admin, education and health           0.0181740  0.0599948
## INDE07MOther services                              -0.0073582  0.0626015
## NSECMJ20Lower managerial and professional          -0.2150006  0.0110268
## NSECMJ20Intermediate occupations                   -0.4471331  0.0139022
## NSECMJ20Lower supervisory and technical            -0.4203188  0.0195287
## NSECMJ20Semi-routine occupations                   -0.5387646  0.0169187
## NSECMJ20Routine occupations                        -0.5727283  0.0177126
## NSECMJ20Never worked                               -0.4505182  0.0307417
## GOR9DYorkshire and The Humber                      -0.1609020  0.0191250
## GOR9DSouth West                                    -0.1249406  0.0191410
## GOR9DWest Midlands                                 -0.1123784  0.0205305
## GOR9DSouth East                                    -0.0892086  0.0176602
## GOR9DEast of England                               -0.0759118  0.0200910
## GOR9DEast Midlands                                 -0.1569061  0.0198246
## GOR9DNorth East                                    -0.1842434  0.0247838
## GOR9DNorth West                                    -0.1410464  0.0195823
## GOR9DWales                                         -0.1764938  0.0250278
## GOR9DNorthern Ireland                              -0.1800145  0.0214698
## GOR9DScotland                                      -0.1140573  0.0199441
##                                                    t value Pr(>|t|)
## (Intercept)                                         40.443  < 2e-16 ***
## SEXFemale                                           -8.092 6.85e-16 ***
## AGE                                                  9.594  < 2e-16 ***
## HIQUL22DHigher education                            -8.534  < 2e-16 ***
## HIQUL22DA level or equivalent                      -15.217  < 2e-16 ***
## HIQUL22DGCSE A*-C or equivalent                    -15.422  < 2e-16 ***
## HIQUL22DOther qualitfication                        -9.424  < 2e-16 ***
## HIQUL22DNo Qualification                           -11.114  < 2e-16 ***
## FLEX22W6No                                           1.918 0.055154 .
## FTPTPart-time                                       -2.741 0.006138 **
## EMPLENThree months, less than 6                     -0.471 0.637953
## EMPLEN6 months, less than 12                         0.497 0.619555
## EMPLEN1 year, less than 2                            0.242 0.808501
## EMPLEN2 years, less than 5                           1.846 0.064879 .
## EMPLEN5 years, less than 10                          2.432 0.015031 *
## EMPLEN10 years, less than 20                         3.291 0.001002 **
## EMPLEN20 years or more                               4.222 2.45e-05 ***
## JOBTYPNot permanent in some way                      0.443 0.657738
## MPNRO211-19                                          0.632 0.527174
## MPNRO220-24                                          0.784 0.433282
## MPNRO2Don't know but under 25                       -0.878 0.380042
## MPNRO225-49                                          0.908 0.363938
## MPNRO250-249                                         3.334 0.000862 ***
## MPNRO2250-499                                        4.733 2.26e-06 ***
## MPNRO2Don't know but between 50 and 499              1.664 0.096159 .
## MPNRO2500 or more                                    8.886  < 2e-16 ***
## MARSTAMarried, living with spouse                    5.989 2.21e-09 ***
```

```
## MARSTAMarried separated from spouse                    1.294 0.195781
## MARSTADivorced                                         3.904 9.56e-05 ***
## MARSTAWidowed                                          0.635 0.525652
## MARSTACurrently or previously in civil partnership     0.032 0.974259
## INDE07MEnergy and water                                2.880 0.003988 **
## INDE07MManufacturing                                   2.002 0.045319 *
## INDE07MConstruction                                    3.128 0.001767 **
## INDE07MDistribution, hotels and restaurants           -0.121 0.903413
## INDE07MTransport and communication                     2.794 0.005218 **
## INDE07MBanking and finance                             2.043 0.041066 *
## INDE07MPublic admin, education and health              0.303 0.761955
## INDE07MOther services                                 -0.118 0.906435
## NSECMJ20Lower managerial and professional            -19.498  < 2e-16 ***
## NSECMJ20Intermediate occupations                     -32.163  < 2e-16 ***
## NSECMJ20Lower supervisory and technical              -21.523  < 2e-16 ***
## NSECMJ20Semi-routine occupations                     -31.844  < 2e-16 ***
## NSECMJ20Routine occupations                          -32.334  < 2e-16 ***
## NSECMJ20Never worked                                 -14.655  < 2e-16 ***
## GOR9DYorkshire and The Humber                         -8.413  < 2e-16 ***
## GOR9DSouth West                                       -6.527 7.14e-11 ***
## GOR9DWest Midlands                                    -5.474 4.55e-08 ***
## GOR9DSouth East                                       -5.051 4.49e-07 ***
## GOR9DEast of England                                  -3.778 0.000159 ***
## GOR9DEast Midlands                                    -7.915 2.85e-15 ***
## GOR9DNorth East                                       -7.434 1.17e-13 ***
## GOR9DNorth West                                       -7.203 6.49e-13 ***
## GOR9DWales                                            -7.052 1.93e-12 ***
## GOR9DNorthern Ireland                                 -8.385  < 2e-16 ***
## GOR9DScotland                                         -5.719 1.11e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3402 on 7291 degrees of freedom
## Multiple R-squared:  0.483,  Adjusted R-squared:  0.4791
## F-statistic: 123.9 on 55 and 7291 DF,  p-value: < 2.2e-16
```

残差与拟合值图



残差QQ图



RESET检验

```
lfs %>%
  mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 +
MARSTA + INDE07M + NSECMJ20 + GOR9D, data=.)  %>%
  resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 52.408, df1 = 2, df2 = 7289, p-value < 2.2e-16
```

BP检验

```
lfs %>%
  mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 +
MARSTA + INDE07M + NSECMJ20 + GOR9D, data=.)  %>%
  bptest()
```

```
##
##  studentized Breusch-Pagan test
##
## data:  .
## BP = 356.74, df = 55, p-value < 2.2e-16
```

一些思考

- 半对数模型带来了显著改善。
- 但它在形式上仍未完全满足模型假设。
- 如果不增加其他变量，这可能已经是能达到的最好状态。
- 好在普通线性模型对一定程度的假设偏离具有较强稳健性。
- 我认为可以主张我们的结果仍比较稳健。
- 接下来看看还能否进一步改进和完善。

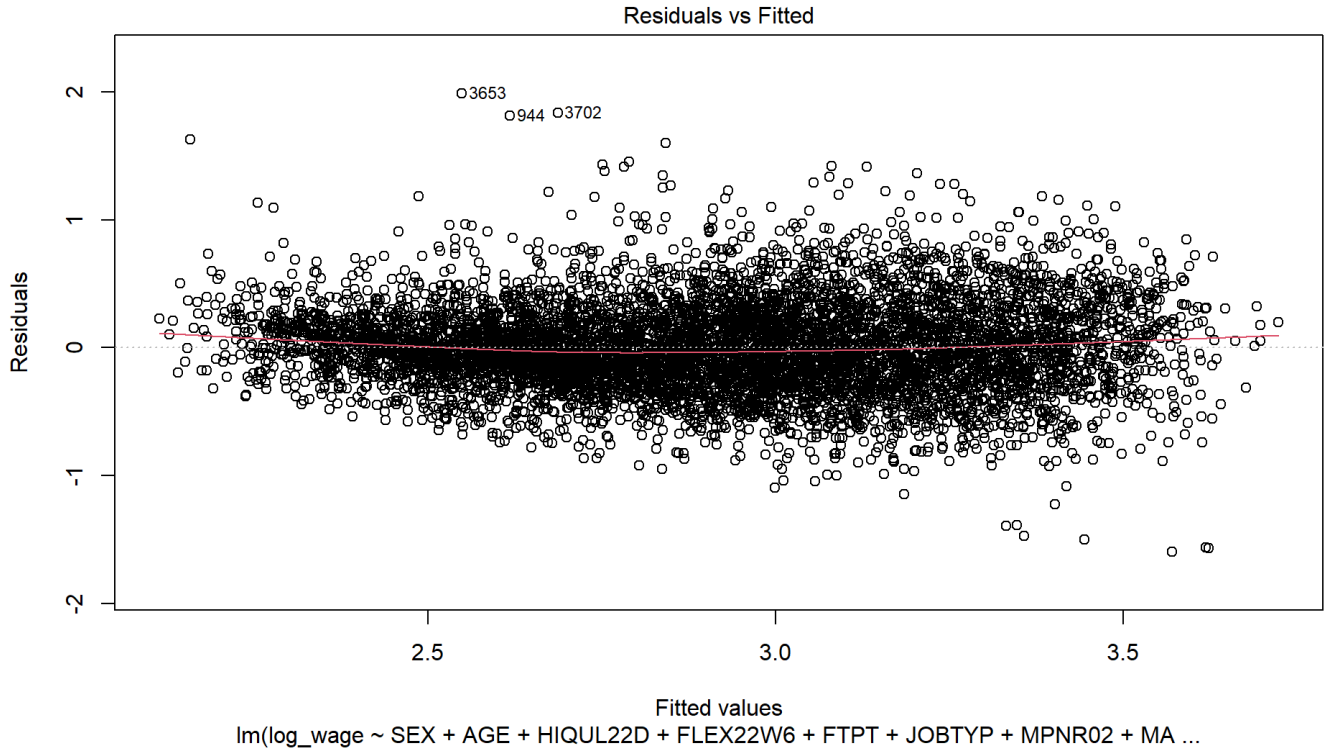
删掉内生变量EMPLEN

```
##
## Call:
## lm(formula = log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT +
##     JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 + GOR9D, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.59585 -0.21204 -0.01958  0.19681  1.99143
##
## Coefficients:
##                                                    Estimate Std. Error
## (Intercept)                                       3.0076449  0.0705405
## SEXFemale                                        -0.0729905  0.0089255
## AGE                                               0.0053555  0.0003982
## HIQUL22DHigher education                         -0.1304069  0.0158969
## HIQUL22DA level or equivalent                    -0.1641653  0.0111481
## HIQUL22DGCSE A*-C or equivalent                  -0.1955231  0.0130727
## HIQUL22DOther qualitfication                     -0.1946198  0.0210410
## HIQUL22DNo Qualification                         -0.2776273  0.0259223
## FLEX22W6No                                        0.0639224  0.0296268
## FTPTPart-time                                    -0.0293177  0.0109014
## JOBTYPNot permanent in some way                  -0.0186073  0.0233054
## MPNR0211-19                                       0.0116080  0.0176365
## MPNR0220-24                                       0.0142293  0.0229066
## MPNR02Don't know but under 25                    -0.0286165  0.0297474
## MPNR0225-49                                       0.0139333  0.0154756
## MPNR0250-249                                      0.0469975  0.0132730
## MPNR02250-499                                     0.0925272  0.0188567
## MPNR02Don't know but between 50 and 499           0.0343483  0.0220521
## MPNR02500 or more                                 0.1278302  0.0134699
## MARSTAMarried, living with spouse                 0.0582442  0.0096969
## MARSTAMarried separated from spouse               0.0292286  0.0278073
## MARSTADivorced                                    0.0625229  0.0173478
## MARSTAWidowed                                     0.0242070  0.0411645
## MARSTACurrently or previously in civil partnership  0.0003256  0.0397911
## INDE07MEnergy and water                           0.1877995  0.0657630
## INDE07MManufacturing                              0.1205071  0.0612521
## INDE07MConstruction                               0.1921047  0.0625760
## INDE07MDistribution, hotels and restaurants      -0.0113958  0.0606417
## INDE07MTransport and communication                0.1670971  0.0611673
## INDE07MBanking and finance                        0.1187895  0.0605080
## INDE07MPublic admin, education and health         0.0183950  0.0602299
## INDE07MOther services                            -0.0086363  0.0628521
## NSECMJ20Lower managerial and professional        -0.2147182  0.0110710
## NSECMJ20Intermediate occupations                 -0.4528381  0.0139338
## NSECMJ20Lower supervisory and technical          -0.4242265  0.0195929
## NSECMJ20Semi-routine occupations                 -0.5515696  0.0169109
## NSECMJ20Routine occupations                      -0.5873253  0.0176843
```

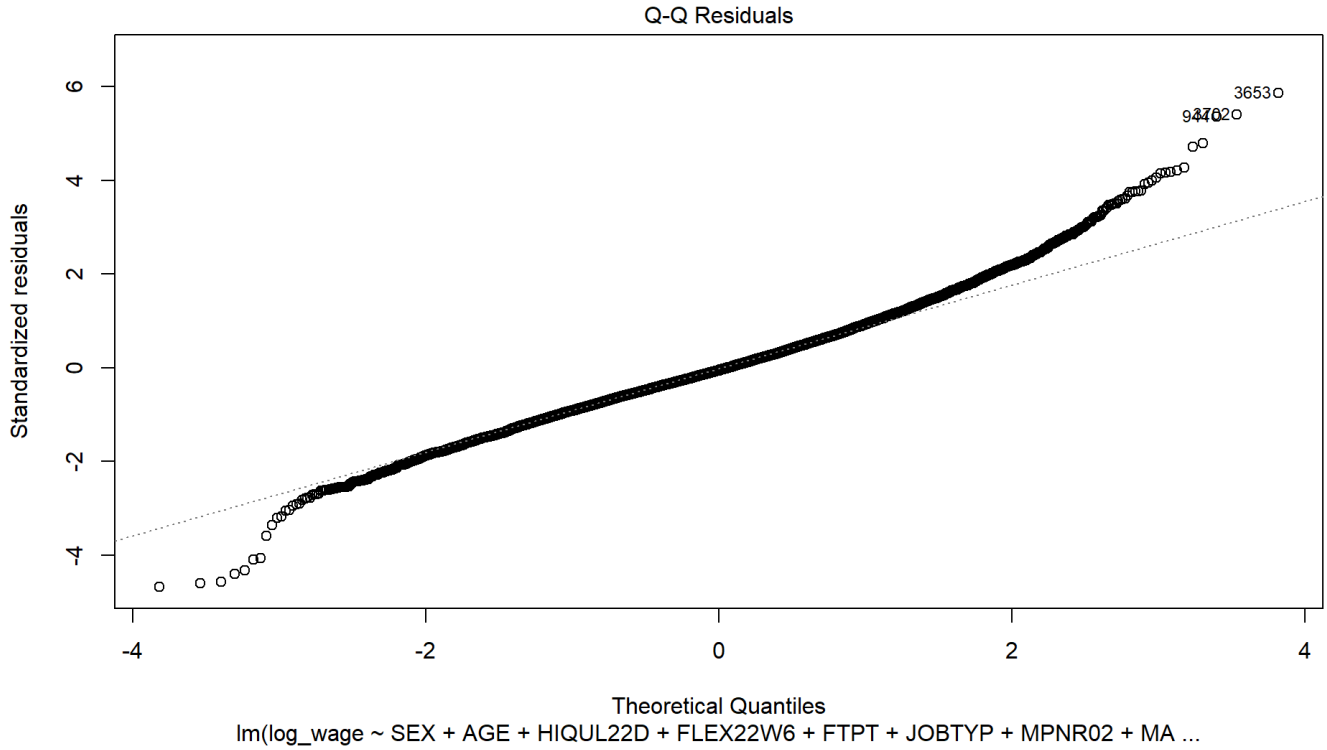
```
## NSECMJ2ONever worked                                -0.4561495  0.0308275
## GOR9DYorkshire and The Humber                       -0.1555679  0.0191598
## GOR9DSouth West                                     -0.1233810  0.0192114
## GOR9DWest Midlands                                  -0.1106066  0.0205938
## GOR9DSouth East                                     -0.0881292  0.0177121
## GOR9DEast of England                                -0.0741064  0.0201603
## GOR9DEast Midlands                                  -0.1528215  0.0198923
## GOR9DNorth East                                     -0.1807970  0.0248637
## GOR9DNorth West                                     -0.1378526  0.0196408
## GOR9DWales                                          -0.1702925  0.0251085
## GOR9DNorthern Ireland                               -0.1665567  0.0214729
## GOR9DScotland                                       -0.1097664  0.0199951
##                                                     t value Pr(>|t|)
## (Intercept)                                          42.637  < 2e-16 ***
## SEXFemale                                            -8.178 3.38e-16 ***
## AGE                                                  13.449  < 2e-16 ***
## HIQUL22DHigher education                             -8.203 2.74e-16 ***
## HIQUL22DA level or equivalent                       -14.726  < 2e-16 ***
## HIQUL22DGCSE A*-C or equivalent                     -14.957  < 2e-16 ***
## HIQUL22DOther qualitfication                         -9.250  < 2e-16 ***
## HIQUL22DNo Qualification                            -10.710  < 2e-16 ***
## FLEX22W6No                                           2.158 0.030992 *
## FTPTPart-time                                       -2.689 0.007175 **
## JOBTYPNot permanent in some way                     -0.798 0.424658
## MPNRO211-19                                          0.658 0.510442
## MPNRO220-24                                          0.621 0.534496
## MPNRO2Don't know but under 25                       -0.962 0.336090
## MPNRO225-49                                          0.900 0.367969
## MPNRO250-249                                         3.541 0.000401 ***
## MPNRO2250-499                                        4.907 9.46e-07 ***
## MPNRO2Don't know but between 50 and 499              1.558 0.119372
## MPNRO2500 or more                                    9.490  < 2e-16 ***
## MARSTAMarried, living with spouse                    6.006 1.99e-09 ***
## MARSTAMarried separated from spouse                  1.051 0.293242
## MARSTADivorced                                       3.604 0.000315 ***
## MARSTAWidowed                                        0.588 0.556514
## MARSTACurrently or previously in civil partnership   0.008 0.993471
## INDE07MEnergy and water                              2.856 0.004306 **
## INDE07MManufacturing                                 1.967 0.049176 *
## INDE07MConstruction                                  3.070 0.002149 **
## INDE07MDistribution, hotels and restaurants         -0.188 0.850945
## INDE07MTransport and communication                   2.732 0.006314 **
## INDE07MBanking and finance                           1.963 0.049660 *
## INDE07MPublic admin, education and health            0.305 0.760060
## INDE07MOther services                               -0.137 0.890714
## NSECMJ2OLower managerial and professional           -19.395  < 2e-16 ***
## NSECMJ2OIntermediate occupations                    -32.499  < 2e-16 ***
## NSECMJ2OLower supervisory and technical             -21.652  < 2e-16 ***
```

```
## NSECMJ20Semi-routine occupations                -32.616  < 2e-16 ***
## NSECMJ20Routine occupations                     -33.212  < 2e-16 ***
## NSECMJ20Never worked                            -14.797  < 2e-16 ***
## GOR9DYorkshire and The Humber                    -8.120 5.45e-16 ***
## GOR9DSouth West                                  -6.422 1.43e-10 ***
## GOR9DWest Midlands                               -5.371 8.08e-08 ***
## GOR9DSouth East                                  -4.976 6.65e-07 ***
## GOR9DEast of England                             -3.676 0.000239 ***
## GOR9DEast Midlands                               -7.682 1.76e-14 ***
## GOR9DNorth East                                  -7.272 3.92e-13 ***
## GOR9DNorth West                                  -7.019 2.44e-12 ***
## GOR9DWales                                       -6.782 1.28e-11 ***
## GOR9DNorthern Ireland                            -7.757 9.91e-15 ***
## GOR9DScotland                                    -5.490 4.16e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3416 on 7298 degrees of freedom
## Multiple R-squared:  0.4782, Adjusted R-squared:  0.4748
## F-statistic: 139.4 on 48 and 7298 DF,  p-value: < 2.2e-16
```

残差与拟合值图



残差QQ图



RESET检验

```
lfs %>%
  mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + JOBTYP + MPNR02 + MARSTA +
INDE07M + NSECMJ20 + GOR9D, data=.)  %>%
  resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 52.253, df1 = 2, df2 = 7296, p-value < 2.2e-16
```

BP检验

```
lfs %>%
  mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT + JOBTYP + MPNR02 + MARSTA +
INDE07M + NSECMJ20 + GOR9D, data=.)  %>%
  bptest()
```

```
##
##  studentized Breusch-Pagan test
##
## data:  .
## BP = 341.84, df = 48, p-value < 2.2e-16
```

当前模型

```
##
## Call:
## lm(formula = log_wage ~ SEX + AGE + HIQUL22D + FLEX22W6 + FTPT +
##     JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 + GOR9D, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.59585 -0.21204 -0.01958  0.19681  1.99143
##
## Coefficients:
##                                                    Estimate Std. Error
## (Intercept)                                       3.0076449  0.0705405
## SEXFemale                                        -0.0729905  0.0089255
## AGE                                               0.0053555  0.0003982
## HIQUL22DHigher education                         -0.1304069  0.0158969
## HIQUL22DA level or equivalent                    -0.1641653  0.0111481
## HIQUL22DGCSE A*-C or equivalent                  -0.1955231  0.0130727
## HIQUL22DOther qualitfication                     -0.1946198  0.0210410
## HIQUL22DNo Qualification                         -0.2776273  0.0259223
## FLEX22W6No                                        0.0639224  0.0296268
## FTPTPart-time                                    -0.0293177  0.0109014
## JOBTYPNot permanent in some way                  -0.0186073  0.0233054
## MPNR0211-19                                       0.0116080  0.0176365
## MPNR0220-24                                       0.0142293  0.0229066
## MPNR02Don't know but under 25                    -0.0286165  0.0297474
## MPNR0225-49                                       0.0139333  0.0154756
## MPNR0250-249                                      0.0469975  0.0132730
## MPNR02250-499                                     0.0925272  0.0188567
## MPNR02Don't know but between 50 and 499           0.0343483  0.0220521
## MPNR02500 or more                                 0.1278302  0.0134699
## MARSTAMarried, living with spouse                 0.0582442  0.0096969
## MARSTAMarried separated from spouse               0.0292286  0.0278073
## MARSTADivorced                                    0.0625229  0.0173478
## MARSTAWidowed                                     0.0242070  0.0411645
## MARSTACurrently or previously in civil partnership  0.0003256  0.0397911
## INDE07MEnergy and water                           0.1877995  0.0657630
## INDE07MManufacturing                              0.1205071  0.0612521
## INDE07MConstruction                               0.1921047  0.0625760
## INDE07MDistribution, hotels and restaurants      -0.0113958  0.0606417
## INDE07MTransport and communication                0.1670971  0.0611673
## INDE07MBanking and finance                        0.1187895  0.0605080
## INDE07MPublic admin, education and health         0.0183950  0.0602299
## INDE07MOther services                            -0.0086363  0.0628521
## NSECMJ20Lower managerial and professional        -0.2147182  0.0110710
## NSECMJ20Intermediate occupations                 -0.4528381  0.0139338
## NSECMJ20Lower supervisory and technical          -0.4242265  0.0195929
## NSECMJ20Semi-routine occupations                 -0.5515696  0.0169109
## NSECMJ20Routine occupations                      -0.5873253  0.0176843
```

```
## NSECMJ2ONever worked                                  -0.4561495  0.0308275
## GOR9DYorkshire and The Humber                         -0.1555679  0.0191598
## GOR9DSouth West                                       -0.1233810  0.0192114
## GOR9DWest Midlands                                    -0.1106066  0.0205938
## GOR9DSouth East                                       -0.0881292  0.0177121
## GOR9DEast of England                                  -0.0741064  0.0201603
## GOR9DEast Midlands                                    -0.1528215  0.0198923
## GOR9DNorth East                                       -0.1807970  0.0248637
## GOR9DNorth West                                       -0.1378526  0.0196408
## GOR9DWales                                            -0.1702925  0.0251085
## GOR9DNorthern Ireland                                 -0.1665567  0.0214729
## GOR9DScotland                                         -0.1097664  0.0199951
##                                                       t value Pr(>|t|)
## (Intercept)                                            42.637  < 2e-16 ***
## SEXFemale                                              -8.178 3.38e-16 ***
## AGE                                                    13.449  < 2e-16 ***
## HIQUL22DHigher education                               -8.203 2.74e-16 ***
## HIQUL22DA level or equivalent                         -14.726  < 2e-16 ***
## HIQUL22DGCSE A*-C or equivalent                       -14.957  < 2e-16 ***
## HIQUL22DOther qualitfication                           -9.250  < 2e-16 ***
## HIQUL22DNo Qualification                              -10.710  < 2e-16 ***
## FLEX22W6No                                             2.158 0.030992 *
## FTPTPart-time                                         -2.689 0.007175 **
## JOBTYPNot permanent in some way                       -0.798 0.424658
## MPNRO211-19                                            0.658 0.510442
## MPNRO220-24                                            0.621 0.534496
## MPNRO2Don't know but under 25                         -0.962 0.336090
## MPNRO225-49                                            0.900 0.367969
## MPNRO250-249                                           3.541 0.000401 ***
## MPNRO2250-499                                          4.907 9.46e-07 ***
## MPNRO2Don't know but between 50 and 499                1.558 0.119372
## MPNRO2500 or more                                      9.490  < 2e-16 ***
## MARSTAMarried, living with spouse                      6.006 1.99e-09 ***
## MARSTAMarried separated from spouse                    1.051 0.293242
## MARSTADivorced                                         3.604 0.000315 ***
## MARSTAWidowed                                          0.588 0.556514
## MARSTACurrently or previously in civil partnership     0.008 0.993471
## INDE07MEnergy and water                                2.856 0.004306 **
## INDE07MManufacturing                                   1.967 0.049176 *
## INDE07MConstruction                                    3.070 0.002149 **
## INDE07MDistribution, hotels and restaurants           -0.188 0.850945
## INDE07MTransport and communication                     2.732 0.006314 **
## INDE07MBanking and finance                             1.963 0.049660 *
## INDE07MPublic admin, education and health              0.305 0.760060
## INDE07MOther services                                 -0.137 0.890714
## NSECMJ2OLower managerial and professional             -19.395  < 2e-16 ***
## NSECMJ2OIntermediate occupations                      -32.499  < 2e-16 ***
## NSECMJ2OLower supervisory and technical               -21.652  < 2e-16 ***
```

```
## NSECMJ20Semi-routine occupations                -32.616  < 2e-16 ***
## NSECMJ20Routine occupations                     -33.212  < 2e-16 ***
## NSECMJ20Never worked                            -14.797  < 2e-16 ***
## GOR9DYorkshire and The Humber                    -8.120 5.45e-16 ***
## GOR9DSouth West                                  -6.422 1.43e-10 ***
## GOR9DWest Midlands                               -5.371 8.08e-08 ***
## GOR9DSouth East                                  -4.976 6.65e-07 ***
## GOR9DEast of England                             -3.676 0.000239 ***
## GOR9DEast Midlands                               -7.682 1.76e-14 ***
## GOR9DNorth East                                  -7.272 3.92e-13 ***
## GOR9DNorth West                                  -7.019 2.44e-12 ***
## GOR9DWales                                       -6.782 1.28e-11 ***
## GOR9DNorthern Ireland                            -7.757 9.91e-15 ***
## GOR9DScotland                                    -5.490 4.16e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3416 on 7298 degrees of freedom
## Multiple R-squared:  0.4782, Adjusted R-squared:  0.4748
## F-statistic: 139.4 on 48 and 7298 DF,  p-value: < 2.2e-16
```

兼职工作与零工时合同

- 模型中的零工时合同变量在统计上不显著。
- 纳入它并没有特别充分的理由，只是认为它可能反映工作质量的某个方面。
- 既然不显著，我们可以删除它。
- 这不太可能对模型造成很大影响。

改进后的模型

```
##
## Call:
## lm(formula = log_wage ~ SEX + AGE + HIQUL22D + FTPT + JOBTYP +
##     MPNRO2 + MARSTA + INDEO7M + NSECMJ2O + GOR9D, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.59664 -0.21275 -0.01911  0.19687  1.99620
##
## Coefficients:
##                                                    Estimate Std. Error
## (Intercept)                                       3.0697098  0.0644249
## SEXFemale                                        -0.0728093  0.0089273
## AGE                                               0.0053838  0.0003981
## HIQUL22DHigher education                         -0.1308017  0.0158998
## HIQUL22DA level or equivalent                    -0.1639414  0.0111504
## HIQUL22DGCSE A*-C or equivalent                  -0.1947025  0.0130705
## HIQUL22DOther qualitfication                     -0.1938950  0.0210435
## HIQUL22DNo Qualification                         -0.2763608  0.0259222
## FTPTPart-time                                    -0.0312090  0.0108688
## JOBTYPNot permanent in some way                  -0.0285256  0.0228532
## MPNRO211-19                                       0.0115431  0.0176409
## MPNRO220-24                                       0.0139960  0.0229121
## MPNRO2Don't know but under 25                    -0.0295990  0.0297514
## MPNRO225-49                                       0.0139838  0.0154794
## MPNRO250-249                                      0.0473970  0.0132750
## MPNRO2250-499                                     0.0934217  0.0188569
## MPNRO2Don't know but between 50 and 499           0.0333879  0.0220531
## MPNRO2500 or more                                 0.1287248  0.0134669
## MARSTAMarried, living with spouse                 0.0585553  0.0096983
## MARSTAMarried separated from spouse               0.0284295  0.0278118
## MARSTADivorced                                    0.0624700  0.0173522
## MARSTAWidowed                                     0.0246423  0.0411744
## MARSTACurrently or previously in civil partnership -0.0001658  0.0398004
## INDEO7MEnergy and water                           0.1880907  0.0657794
## INDEO7MManufacturing                              0.1211473  0.0612667
## INDEO7MConstruction                               0.1931411  0.0625899
## INDEO7MDistribution, hotels and restaurants      -0.0119765  0.0606563
## INDEO7MTransport and communication                0.1677362  0.0611819
## INDEO7MBanking and finance                        0.1194270  0.0605224
## INDEO7MPublic admin, education and health         0.0187608  0.0602447
## INDEO7MOther services                            -0.0089625  0.0628677
## NSECMJ2OLower managerial and professional        -0.2148127  0.0110737
## NSECMJ2OIntermediate occupations                 -0.4534746  0.0139341
## NSECMJ2OLower supervisory and technical          -0.4249974  0.0195945
## NSECMJ2OSemi-routine occupations                 -0.5532582  0.0168970
## NSECMJ2ORoutine occupations                      -0.5906754  0.0176204
## NSECMJ2ONever worked                             -0.4637241  0.0306346
```

```
## GOR9DYorkshire and The Humber                          -0.1552762  0.0191641
## GOR9DSouth West                                        -0.1234761  0.0192162
## GOR9DWest Midlands                                     -0.1107999  0.0205987
## GOR9DSouth East                                        -0.0888658  0.0177132
## GOR9DEast of England                                   -0.0750130  0.0201609
## GOR9DEast Midlands                                     -0.1529863  0.0198972
## GOR9DNorth East                                        -0.1809091  0.0248699
## GOR9DNorth West                                        -0.1377077  0.0196456
## GOR9DWales                                             -0.1703914  0.0251148
## GOR9DNorthern Ireland                                  -0.1663384  0.0214780
## GOR9DScotland                                          -0.1102690  0.0199988
##                                                        t value Pr(>|t|)
## (Intercept)                                             47.648  < 2e-16 ***
## SEXFemale                                               -8.156 4.05e-16 ***
## AGE                                                     13.524  < 2e-16 ***
## HIQUL22DHigher education                                -8.227 2.26e-16 ***
## HIQUL22DA level or equivalent                          -14.703  < 2e-16 ***
## HIQUL22DGCSE A*-C or equivalent                        -14.896  < 2e-16 ***
## HIQUL22DOther qualitfication                            -9.214  < 2e-16 ***
## HIQUL22DNo Qualification                               -10.661  < 2e-16 ***
## FTPTPart-time                                           -2.871 0.004098 **
## JOBTYPNot permanent in some way                         -1.248 0.211995
## MPNRO211-19                                              0.654 0.512914
## MPNRO220-24                                              0.611 0.541315
## MPNRO2Don't know but under 25                           -0.995 0.319828
## MPNRO225-49                                              0.903 0.366355
## MPNRO250-249                                             3.570 0.000359 ***
## MPNRO2250-499                                            4.954 7.42e-07 ***
## MPNRO2Don't know but between 50 and 499                  1.514 0.130076
## MPNRO2500 or more                                        9.559  < 2e-16 ***
## MARSTAMarried, living with spouse                        6.038 1.64e-09 ***
## MARSTAMarried separated from spouse                      1.022 0.306715
## MARSTADivorced                                           3.600 0.000320 ***
## MARSTAWidowed                                            0.598 0.549533
## MARSTACurrently or previously in civil partnership      -0.004 0.996677
## INDEO7MEnergy and water                                  2.859 0.004256 **
## INDEO7MManufacturing                                     1.977 0.048037 *
## INDEO7MConstruction                                      3.086 0.002038 **
## INDEO7MDistribution, hotels and restaurants             -0.197 0.843482
## INDEO7MTransport and communication                       2.742 0.006129 **
## INDEO7MBanking and finance                               1.973 0.048503 *
## INDEO7MPublic admin, education and health                0.311 0.755498
## INDEO7MOther services                                   -0.143 0.886641
## NSECMJ2OLower managerial and professional              -19.399  < 2e-16 ***
## NSECMJ2OIntermediate occupations                       -32.544  < 2e-16 ***
## NSECMJ2OLower supervisory and technical                -21.690  < 2e-16 ***
## NSECMJ2OSemi-routine occupations                       -32.743  < 2e-16 ***
## NSECMJ2ORoutine occupations                            -33.522  < 2e-16 ***
```

```
## NSECMJ20Never worked                                  -15.137  < 2e-16 ***
## GOR9DYorkshire and The Humber                          -8.102 6.27e-16 ***
## GOR9DSouth West                                        -6.426 1.40e-10 ***
## GOR9DWest Midlands                                     -5.379 7.72e-08 ***
## GOR9DSouth East                                        -5.017 5.37e-07 ***
## GOR9DEast of England                                   -3.721 0.000200 ***
## GOR9DEast Midlands                                     -7.689 1.68e-14 ***
## GOR9DNorth East                                        -7.274 3.85e-13 ***
## GOR9DNorth West                                        -7.010 2.60e-12 ***
## GOR9DWales                                             -6.785 1.26e-11 ***
## GOR9DNorthern Ireland                                  -7.745 1.09e-14 ***
## GOR9DScotland                                          -5.514 3.63e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3417 on 7299 degrees of freedom
## Multiple R-squared:  0.4779, Adjusted R-squared:  0.4745
## F-statistic: 142.1 on 47 and 7299 DF,  p-value: < 2.2e-16
```

婚姻状况

- 婚姻状况似乎很重要，但只有两种类别显著：已婚且与配偶同住，以及离婚。
- 可以考虑把它减少为两个类别。
- 有多种实现方法。
- 我打算使用case_when()函数。
- 随后重新运行回归。

创建新变量

```
lfs <- lfs %>%
  mutate(new_MARSTA = case_when(
    MARSTA == "Married, living with spouse" ~ "Married",
    MARSTA == "Divorced" ~ "Divorced",
    TRUE ~ "Other") %>%
      factor() %>%
      relevel(ref = "Other")
    )
```

更改行业变量的参照组

- 我们发现行业变量的参照类别是规模最小的行业。
- 将规模最大的行业改为参照类别。

```
lfs <- lfs %>%
  mutate(INDE07M = relevel(INDE07M, ref = "Public admin, education and health"))
```

将年龄改为分类变量

- 为了允许更多非线性效应，可以使用数据集已有的5岁年龄组。
- 需要修改现有脚本，加入AGES变量。
- 也可以使用更细的分组，但目前的分组已经相当详细。

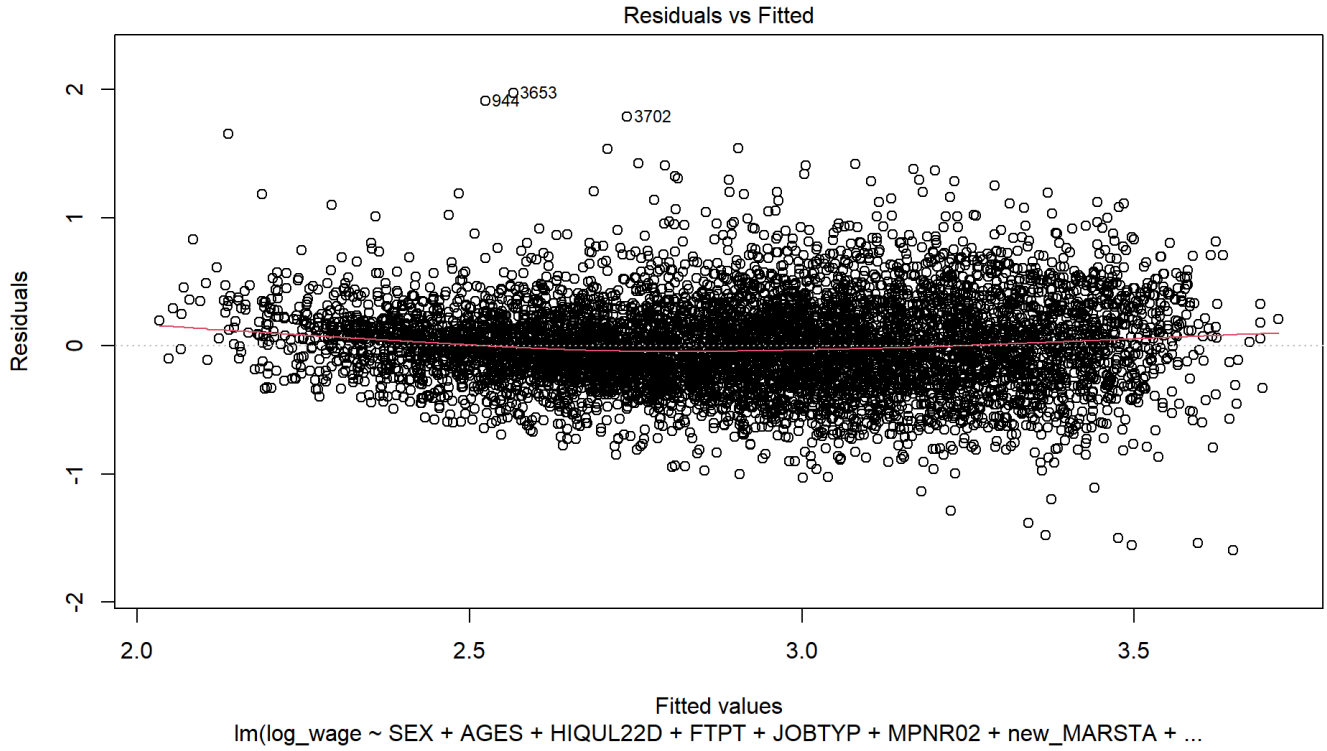
改进后的模型

```
##
## Call:
## lm(formula = log_wage ~ SEX + AGES + HIQUL22D + FTPT + JOBTYP +
##     MPNRO2 + new_MARSTA + INDEO7M + NSECMJ2O + GOR9D, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.59477 -0.21330 -0.01482  0.19322  1.97498
##
## Coefficients:
##                                              Estimate Std. Error t value
## (Intercept)                                  3.071377   0.040801  75.276
## SEXFemale                                   -0.077926   0.008856  -8.799
## AGES20-24 years                              0.029881   0.037176   0.804
## AGES25-29 years                              0.113455   0.036701   3.091
## AGES30-34 years                              0.210264   0.036244   5.801
## AGES35-39 years                              0.269129   0.036544   7.365
## AGES40-44 years                              0.293082   0.036508   8.028
## AGES45-49 years                              0.319701   0.036825   8.682
## AGES50-54 years                              0.316385   0.036721   8.616
## AGES55-59 years                              0.333609   0.036765   9.074
## AGES60-64 years                              0.270392   0.037477   7.215
## HIQUL22DHigher education                    -0.124200   0.015773  -7.874
## HIQUL22DA level or equivalent               -0.153325   0.011123 -13.784
## HIQUL22DGCSE A*-C or equivalent             -0.189458   0.013051 -14.517
## HIQUL22DOther qualitfication                -0.201310   0.020865  -9.648
## HIQUL22DNo Qualification                    -0.275918   0.025699 -10.737
## FTPTPart-time                               -0.025525   0.010832  -2.356
## JOBTYPNot permanent in some way             -0.019869   0.022661  -0.877
## MPNRO211-19                                  0.015408   0.017492   0.881
## MPNRO220-24                                  0.014779   0.022719   0.651
## MPNRO2Don't know but under 25               -0.025067   0.029505  -0.850
## MPNRO225-49                                  0.010905   0.015348   0.711
## MPNRO250-249                                 0.042461   0.013159   3.227
## MPNRO2250-499                                0.090216   0.018693   4.826
## MPNRO2Don't know but between 50 and 499      0.031594   0.021870   1.445
## MPNRO2500 or more                            0.123912   0.013347   9.284
## new_MARSTADivorced                           0.048774   0.016814   2.901
## new_MARSTAMarried                            0.033619   0.009318   3.608
## INDEO7MAgriculture, forestry and fishing    -0.010590   0.059719  -0.177
## INDEO7MEnergy and water                      0.163085   0.028486   5.725
## INDEO7MManufacturing                         0.105105   0.015742   6.677
## INDEO7MConstruction                          0.171071   0.020351   8.406
## INDEO7MDistribution, hotels and restaurants -0.026065   0.013524  -1.927
## INDEO7MTransport and communication           0.144873   0.015278   9.483
## INDEO7MBanking and finance                   0.100284   0.011866   8.451
## INDEO7MOther services                       -0.025212   0.020905  -1.206
## NSECMJ2OLower managerial and professional   -0.213346   0.010974 -19.440
```

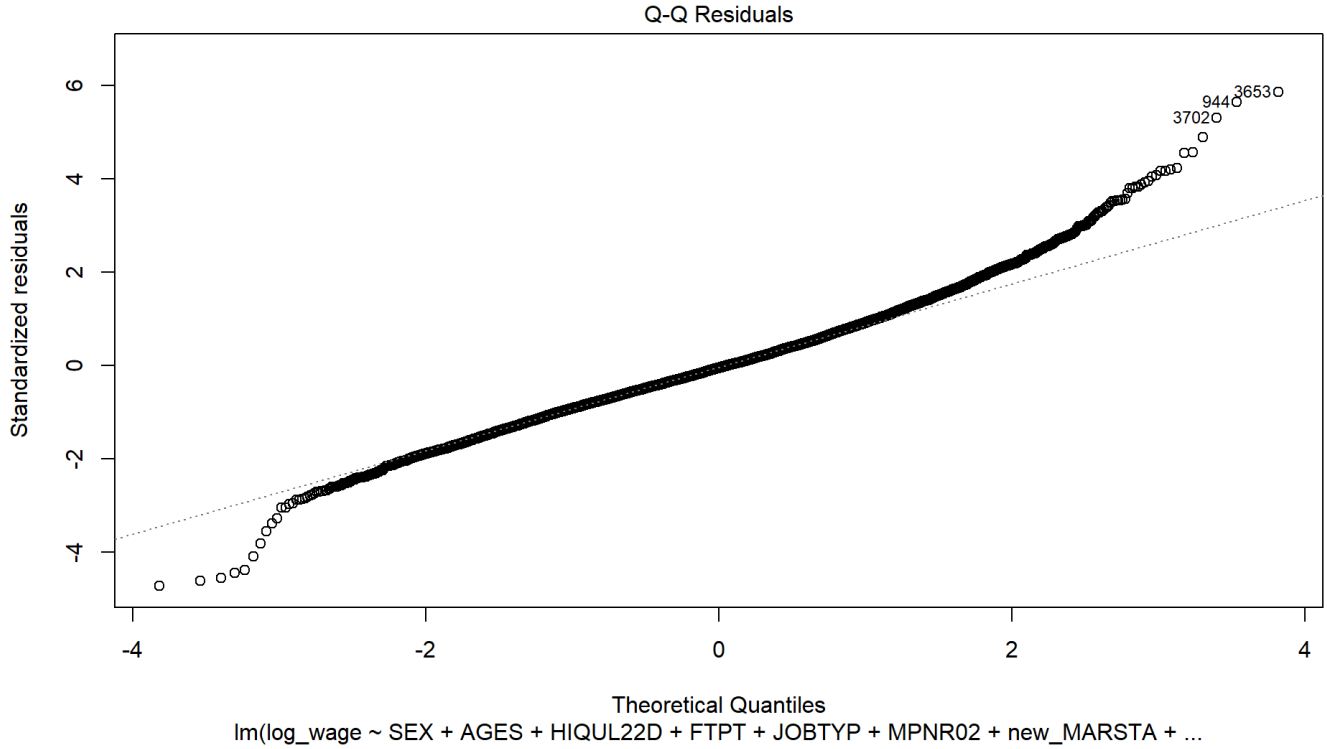
```
## NSECMJ20Intermediate occupations            -0.442134   0.013848 -31.927
## NSECMJ20Lower supervisory and technical     -0.414919   0.019441 -21.342
## NSECMJ20Semi-routine occupations            -0.542399   0.016767 -32.350
## NSECMJ20Routine occupations                 -0.576180   0.017511 -32.903
## NSECMJ20Never worked                        -0.405443   0.032138 -12.616
## GOR9DYorkshire and The Humber               -0.152446   0.018989  -8.028
## GOR9DSouth West                             -0.125708   0.019045  -6.601
## GOR9DWest Midlands                          -0.111728   0.020417  -5.472
## GOR9DSouth East                             -0.083246   0.017551  -4.743
## GOR9DEast of England                        -0.072149   0.019976  -3.612
## GOR9DEast Midlands                          -0.150892   0.019716  -7.653
## GOR9DNorth East                             -0.185774   0.024636  -7.541
## GOR9DNorth West                             -0.137142   0.019472  -7.043
## GOR9DWales                                  -0.172875   0.024898  -6.943
## GOR9DNorthern Ireland                       -0.161764   0.021287  -7.599
## GOR9DScotland                               -0.103908   0.019825  -5.241
##                                            Pr(>|t|)
## (Intercept)                                 < 2e-16 ***
## SEXFemale                                   < 2e-16 ***
## AGES20-24 years                            0.421545
## AGES25-29 years                            0.002000 **
## AGES30-34 years                            6.85e-09 ***
## AGES35-39 years                            1.97e-13 ***
## AGES40-44 years                            1.15e-15 ***
## AGES45-49 years                             < 2e-16 ***
## AGES50-54 years                             < 2e-16 ***
## AGES55-59 years                             < 2e-16 ***
## AGES60-64 years                            5.94e-13 ***
## HIQUL22DHigher education                   3.93e-15 ***
## HIQUL22DA level or equivalent               < 2e-16 ***
## HIQUL22DGCSE A*-C or equivalent             < 2e-16 ***
## HIQUL22DOther qualitfication                < 2e-16 ***
## HIQUL22DNo Qualification                    < 2e-16 ***
## FTPTPart-time                              0.018478 *
## JOBTYPNot permanent in some way            0.380621
## MPNRO211-19                                0.378430
## MPNRO220-24                                0.515380
## MPNRO2Don't know but under 25              0.395579
## MPNRO225-49                                0.477383
## MPNRO250-249                               0.001258 **
## MPNRO2250-499                              1.42e-06 ***
## MPNRO2Don't know but between 50 and 499    0.148611
## MPNRO2500 or more                           < 2e-16 ***
## new_MARSTADivorced                         0.003734 **
## new_MARSTAMarried                          0.000311 ***
## INDE07MAgriculture, forestry and fishing   0.859254
## INDE07MEnergy and water                    1.07e-08 ***
## INDE07MManufacturing                       2.62e-11 ***
```

```
## INDE07MConstruction                              < 2e-16 ***
## INDE07MDistribution, hotels and restaurants 0.053971 .
## INDE07MTransport and communication               < 2e-16 ***
## INDE07MBanking and finance                       < 2e-16 ***
## INDE07MOther services                           0.227851
## NSECMJ20Lower managerial and professional        < 2e-16 ***
## NSECMJ20Intermediate occupations                 < 2e-16 ***
## NSECMJ20Lower supervisory and technical          < 2e-16 ***
## NSECMJ20Semi-routine occupations                 < 2e-16 ***
## NSECMJ20Routine occupations                      < 2e-16 ***
## NSECMJ20Never worked                             < 2e-16 ***
## GOR9DYorkshire and The Humber                   1.14e-15 ***
## GOR9DSouth West                                 4.38e-11 ***
## GOR9DWest Midlands                              4.59e-08 ***
## GOR9DSouth East                                 2.14e-06 ***
## GOR9DEast of England                            0.000306 ***
## GOR9DEast Midlands                              2.21e-14 ***
## GOR9DNorth East                                 5.24e-14 ***
## GOR9DNorth West                                 2.05e-12 ***
## GOR9DWales                                      4.16e-12 ***
## GOR9DNorthern Ireland                           3.35e-14 ***
## GOR9DScotland                                   1.64e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3385 on 7294 degrees of freedom
## Multiple R-squared:  0.4878, Adjusted R-squared:  0.4842
## F-statistic: 133.6 on 52 and 7294 DF,  p-value: < 2.2e-16
```

残差与拟合值图



残差QQ图



RESET检验

```
lfs %>%
  mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGES + HIQUL22D + FTPT + JOBTYP + MPNRO2 + new_MARSTA + INDEO
7M + NSECMJ2O + GOR9D, data=.) %>%
  resettest()
```

```
##
##  RESET test
##
## data:  .
## RESET = 90.25, df1 = 2, df2 = 7292, p-value < 2.2e-16
```

BP检验

```
lfs %>%
  mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGES + HIQUL22D + FTPT + JOBTYP + MPNRO2 + new_MARSTA + INDEO
7M + NSECMJ2O + GOR9D, data=.)  %>%
  bptest()
```

```
##
##  studentized Breusch-Pagan test
##
## data:  .
## BP = 331.69, df = 52, p-value < 2.2e-16
```

异方差

- 改进后模型的异方差问题有所减轻。
- 但仍然存在。
- 我们事先就有理由预期存在异方差。
- 到这个阶段，我建议进行必要的调整。

OLS估计量的推导

- 前面的课程中，我们已经看过OLS估计量推导的基础知识。
- 假设X是解释变量矩阵，y是因变量观测值向量，u是残差向量。
- 可推导出$\beta = (X'X)^{-1}X'y$。
- 还必须推导β参数的标准误。

标准误的推导

- 这里不展开全部推导过程。
- 在推导的某一步，我们得到标准误相关的表达式：
  $E[(\hat{\beta}-\beta)(\hat{\beta}-\beta)']=(X'X)^{-1}X'E[uu']X(X'X)^{-1}$。
- 假设残差方差恒定，即$E[uu']=\sigma^2 I$。
- 这样就能把标准误相关计算简化为$\sigma^2(X'X)^{-1}$。
- 存在异方差时，$E[uu']\neq\sigma^2 I$，因此需要另行说明它的形式。

稳健标准误

- 关于应如何计算标准误，学界提出了多种建议。
- 我们不深入具体细节；实践中，不同方法通常不会造成太大差异。
- 在R中可以采用不同实现方式。
- 我建议结合使用lmtest包中的coeftest()函数和sandwich包中的vcovHC()函数。

稳健标准误

```
lfs %>%
  mutate(log_wage = log(HOURPAY)) %>%
  lm(log_wage ~ SEX + AGES + HIQUL22D + FTPT + JOBTYP + MPNRO2 + new_MARSTA + INDEO
7M + NSECMJ2O + GOR9D, data=.) %>%
coeftest(., vcov = vcovHC(.))
```

```
##
## t test of coefficients:
##
##                                              Estimate Std. Error  t value
## (Intercept)                                 3.0713766  0.0413786  74.2262
## SEXFemale                                  -0.0779263  0.0090922  -8.5707
## AGES20-24 years                             0.0298814  0.0345236   0.8655
## AGES25-29 years                             0.1134549  0.0353139   3.2128
## AGES30-34 years                             0.2102643  0.0351979   5.9738
## AGES35-39 years                             0.2691287  0.0356396   7.5514
## AGES40-44 years                             0.2930818  0.0355250   8.2500
## AGES45-49 years                             0.3197014  0.0358861   8.9088
## AGES50-54 years                             0.3163846  0.0357728   8.8443
## AGES55-59 years                             0.3336090  0.0358905   9.2952
## AGES60-64 years                             0.2703918  0.0367223   7.3632
## HIQUL22DHigher education                   -0.1242005  0.0158922  -7.8152
## HIQUL22DA level or equivalent              -0.1533249  0.0111090 -13.8018
## HIQUL22DGCSE A*-C or equivalent            -0.1894577  0.0126873 -14.9329
## HIQUL22DOther qualitfication               -0.2013098  0.0206791  -9.7349
## HIQUL22DNo Qualification                   -0.2759178  0.0220550 -12.5104
## FTPTPart-time                              -0.0255251  0.0111016  -2.2992
## JOBTYPNot permanent in some way            -0.0198692  0.0236420  -0.8404
## MPNRO211-19                                 0.0154079  0.0184774   0.8339
## MPNRO220-24                                 0.0147793  0.0230963   0.6399
## MPNRO2Don't know but under 25              -0.0250670  0.0251590  -0.9963
## MPNRO225-49                                 0.0109053  0.0155900   0.6995
## MPNRO250-249                                0.0424610  0.0137964   3.0777
## MPNRO2250-499                               0.0902157  0.0190852   4.7270
## MPNRO2Don't know but between 50 and 499     0.0315939  0.0228661   1.3817
## MPNRO2500 or more                           0.1239118  0.0143013   8.6644
## new_MARSTADivorced                          0.0487740  0.0169955   2.8698
## new_MARSTAMarried                           0.0336194  0.0092442   3.6368
## INDE07MAgriculture, forestry and fishing   -0.0105899  0.0647676  -0.1635
## INDE07MEnergy and water                     0.1630855  0.0297111   5.4890
## INDE07MManufacturing                        0.1051048  0.0151844   6.9219
## INDE07MConstruction                         0.1710705  0.0214098   7.9903
## INDE07MDistribution, hotels and restaurants -0.0260650  0.0125052  -2.0843
## INDE07MTransport and communication          0.1448735  0.0167413   8.6536
## INDE07MBanking and finance                  0.1002843  0.0128046   7.8319
## INDE07MOther services                      -0.0252120  0.0221979  -1.1358
## NSECMJ20Lower managerial and professional  -0.2133463  0.0121027 -17.6281
## NSECMJ20Intermediate occupations           -0.4421343  0.0141178 -31.3175
## NSECMJ20Lower supervisory and technical    -0.4149188  0.0195888 -21.1814
## NSECMJ20Semi-routine occupations           -0.5423988  0.0157077 -34.5307
## NSECMJ20Routine occupations                -0.5761800  0.0172005 -33.4980
## NSECMJ20Never worked                       -0.4054428  0.0354811 -11.4270
## GOR9DYorkshire and The Humber              -0.1524458  0.0207421  -7.3496
## GOR9DSouth West                            -0.1257078  0.0207659  -6.0536
```

```
## GOR9DWest Midlands                              -0.1117282  0.0226570  -4.9313
## GOR9DSouth East                                 -0.0832463  0.0201504  -4.1312
## GOR9DEast of England                            -0.0721491  0.0220197  -3.2766
## GOR9DEast Midlands                              -0.1508918  0.0214911  -7.0211
## GOR9DNorth East                                 -0.1857739  0.0237966  -7.8067
## GOR9DNorth West                                 -0.1371423  0.0210648  -6.5105
## GOR9DWales                                      -0.1728748  0.0242134  -7.1396
## GOR9DNorthern Ireland                           -0.1617639  0.0218049  -7.4187
## GOR9DScotland                                   -0.1039077  0.0218131  -4.7636
##                                                  Pr(>|t|)
## (Intercept)                                     < 2.2e-16 ***
## SEXFemale                                       < 2.2e-16 ***
## AGES20-24 years                                  0.386774
## AGES25-29 years                                  0.001320 **
## AGES30-34 years                                 2.427e-09 ***
## AGES35-39 years                                 4.830e-14 ***
## AGES40-44 years                                 < 2.2e-16 ***
## AGES45-49 years                                 < 2.2e-16 ***
## AGES50-54 years                                 < 2.2e-16 ***
## AGES55-59 years                                 < 2.2e-16 ***
## AGES60-64 years                                 1.993e-13 ***
## HIQUL22DHigher education                        6.259e-15 ***
## HIQUL22DA level or equivalent                   < 2.2e-16 ***
## HIQUL22DGCSE A*-C or equivalent                 < 2.2e-16 ***
## HIQUL22DOther qualitfication                    < 2.2e-16 ***
## HIQUL22DNo Qualification                        < 2.2e-16 ***
## FTPTPart-time                                    0.021520 *
## JOBTYPNot permanent in some way                  0.400701
## MPNRO211-19                                      0.404377
## MPNRO220-24                                      0.522259
## MPNRO2Don't know but under 25                    0.319117
## MPNRO225-49                                      0.484256
## MPNRO250-249                                     0.002094 **
## MPNRO2250-499                                   2.321e-06 ***
## MPNRO2Don't know but between 50 and 499          0.167108
## MPNRO2500 or more                               < 2.2e-16 ***
## new_MARSTADivorced                               0.004119 **
## new_MARSTAMarried                                0.000278 ***
## INDE07MAgriculture, forestry and fishing         0.870124
## INDE07MEnergy and water                         4.177e-08 ***
## INDE07MManufacturing                            4.835e-12 ***
## INDE07MConstruction                             1.553e-15 ***
## INDE07MDistribution, hotels and restaurants      0.037165 *
## INDE07MTransport and communication              < 2.2e-16 ***
## INDE07MBanking and finance                      5.486e-15 ***
## INDE07MOther services                            0.256084
## NSECMJ20Lower managerial and professional       < 2.2e-16 ***
## NSECMJ20Intermediate occupations                < 2.2e-16 ***
```

```
## NSECMJ20Lower supervisory and technical     < 2.2e-16 ***
## NSECMJ20Semi-routine occupations            < 2.2e-16 ***
## NSECMJ20Routine occupations                 < 2.2e-16 ***
## NSECMJ20Never worked                        < 2.2e-16 ***
## GOR9DYorkshire and The Humber               2.204e-13 ***
## GOR9DSouth West                             1.487e-09 ***
## GOR9DWest Midlands                          8.350e-07 ***
## GOR9DSouth East                             3.648e-05 ***
## GOR9DEast of England                         0.001056 **
## GOR9DEast Midlands                          2.399e-12 ***
## GOR9DNorth East                             6.688e-15 ***
## GOR9DNorth West                             7.987e-11 ***
## GOR9DWales                                  1.026e-12 ***
## GOR9DNorthern Ireland                       1.316e-13 ***
## GOR9DScotland                               1.939e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

一些思考

- 应检查模型中各变量的统计显著性是否发生变化。
- 现在可能需要进一步排除某些变量。
- 在本例中，这种调整并未造成太大区别。
- 但仍然应当检查并报告稳健标准误，确保结果可信。

检查多重共线性

```
##                GVIF Df GVIF^(1/(2*Df))
## SEX        1.250493  1        1.118255
## AGES       1.923292  9        1.037004
## HIQUL22D   1.635433  5        1.050421
## FTPT       1.272717  1        1.128148
## JOBTYP     1.044482  1        1.021999
## MPNR02     1.256183  8        1.014357
## new_MARSTA 1.361966  2        1.080293
## INDE07M    1.671391  8        1.032624
## NSECMJ20   2.352050  6        1.073875
## GOR9D      1.174285 11        1.007329
```

检查离群值



总结

- 我们的模型比以前更好了。
- 我认为模型大概已经接近能够达到的最好状态。
- 我建议将它视为足够好。
- 下一讲我们会讨论剩余的其他模型假设。